Multiplying and Dividing Fractions: Find the Flip Mistake Answer Key

Grade 4–5

Quick Answers

1. $\frac{1}{4}$
2. $\frac{15}{8}$
3. $\frac{5}{6}$
4. $\frac{32}{45}$
5. $\frac{2}{3}$
6. $\frac{16}{7}$
7. $\frac{4}{3}$
8. 6

Curriculum

Level: Grade 4–5
5.NF.A.1 – *problems 1, 4, 5*
6.NS.A.1 – *problems 2, 3, 6, 7, 8*
Difficulty 1–5 of 5 across 8 tagged checks.

1. Multiply $\frac{2}{3} \times \frac{3}{8}$, cancelling first.
Solution: The 3 on top of the first fraction and the 3 on the bottom of the second are a top-and-bottom pair, so they cancel:

$$\frac{2}{3} \times \frac{3}{8} = \frac{2}{1} \times \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$$

Cancelling is optional here — multiplying straight across gives $\frac{6}{24}$, which reduces to the same thing. $\boxed{\frac{1}{4}}$

Watch for: a student who "keeps, changes, flips" on a multiplication gets $\frac{2}{3} \times \frac{8}{3} = \frac{16}{9}$. Flipping belongs to division only.

2. Divide $\frac{3}{4} \div \frac{2}{5}$.
Solution: The divisor — the fraction *after* the division sign — is the one that flips, because dividing by $\frac{2}{5}$ is the same as multiplying by $\frac{5}{2}$:

$$\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$$

$\boxed{\frac{15}{8}}$ (that is $1\frac{7}{8}$).

Watch for: flipping the first fraction instead gives $\frac{4}{3} \times \frac{2}{5} = \frac{8}{15}$ — less than 1, when the answer must be more than 1 because $\frac{3}{4}$ is bigger than $\frac{2}{5}$.

3. Find and fix Kai's work: $\frac{2}{3} \div \frac{4}{5} = \frac{3}{2} \times \frac{4}{5} = \frac{6}{5}$.
(a) The mistake: Kai flipped the *dividend* $\frac{2}{3}$ (the fraction being divided) instead of the *divisor* $\frac{4}{5}$. Keep, change, flip means keep the FIRST fraction exactly as it is.
(b) Correct work:

$$\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4} = \frac{10}{12} = \frac{5}{6}$$

$\boxed{\frac{5}{6}}$ A quick sanity check: dividing by a number less than 1 makes the result bigger than $\frac{2}{3}$, and $\frac{5}{6}$ is.

4. Find and fix Maya's work: $\frac{4}{5} \times \frac{8}{9} = \frac{1}{5} \times \frac{2}{9} = \frac{2}{45}$.
(a) The mistake: the 4 and the 8 are both on TOP. Cancelling removes the same factor from a numerator and a denominator, which keeps the value the same; removing a factor of 4 from two numerators divides the product by 16, so the answer comes out far too small.
(b) Correct work: nothing here can cancel (4 and 9 share no factor; 8 and 5 share no factor), so multiply straight across:

$$\frac{4}{5} \times \frac{8}{9} = \frac{32}{45}$$

$\boxed{\frac{32}{45}}$

5. Find the missing factor: $\dfrac{3}{4} \times \underline{\hspace{2cm}} = \dfrac{1}{2}$.

Solution: To undo multiplying by $\dfrac{3}{4}$, divide by $\dfrac{3}{4}$ — that is, multiply by its reciprocal $\dfrac{4}{3}$:

$$\frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \times \frac{4}{3} = \frac{4}{6} = \frac{2}{3}$$

Check: $\dfrac{3}{4} \times \dfrac{2}{3} = \dfrac{6}{12} = \dfrac{1}{2}$. $\boxed{\dfrac{2}{3}}$

6. Find and fix Jordan's work: $\dfrac{6}{7} \div \dfrac{3}{8} = \dfrac{3}{7} \div \dfrac{3}{4} = \dfrac{3}{7} \times \dfrac{4}{3} = \dfrac{4}{7}$.

(a) The mistake: Jordan cross-cancelled the 6 and the 8 while the problem was still a division. Cross-cancelling is a shortcut for multiplication only. Once the division is rewritten as $\dfrac{6}{7} \times \dfrac{8}{3}$, the 6 and the 8 are both on top, so they still cannot cancel with each other.

(b) Correct work:

$$\frac{6}{7} \div \frac{3}{8} = \frac{6}{7} \times \frac{8}{3} = \frac{2}{7} \times \frac{8}{1} = \frac{16}{7}$$

(here the 6 on top and the 3 on the bottom legally cancel by 3). $\boxed{\dfrac{16}{7}}$ (that is $2\frac{2}{7}$).

7. Ali: $\dfrac{6}{5} \times \dfrac{5}{8} = \dfrac{3}{4}$. Sam: $\dfrac{5}{6} \times \dfrac{8}{5} = \dfrac{4}{3}$. Who is right?

Solution: Sam is right. The problem is $\dfrac{5}{6} \div \dfrac{5}{8}$, so the divisor $\dfrac{5}{8}$ is what flips, giving $\dfrac{5}{6} \times \dfrac{8}{5} = \dfrac{40}{30} = \dfrac{4}{3}$. Ali took the reciprocal of the dividend $\dfrac{5}{6}$ instead, which answers a different question. $\boxed{\dfrac{4}{3}}$

Check without computing: $\dfrac{5}{6}$ is larger than $\dfrac{5}{8}$ (same numerator, smaller denominator wins), so the quotient must be greater than 1. Ali's $\dfrac{3}{4}$ is less than 1, so it cannot be right.

8. Granola: $\dfrac{3}{4}$ cup per batch, $\dfrac{9}{2}$ cups of oats. How many batches? Nia wrote $\dfrac{2}{9} \times \dfrac{3}{4} = \dfrac{1}{6}$ batch.

(a) Why Nia cannot be right: she has more oats than one batch needs, so the answer must be greater than 1; $\dfrac{1}{6}$ is less than 1. She flipped the dividend $\dfrac{9}{2}$ instead of the divisor $\dfrac{3}{4}$.

(b) Correct work: "how many $\dfrac{3}{4}$-cup scoops fit in $\dfrac{9}{2}$ cups" is a division, and only the divisor flips:

$$\frac{9}{2} \div \frac{3}{4} = \frac{9}{2} \times \frac{4}{3} = \frac{36}{6} = 6$$

$\boxed{\text{6 batches}}$